

Probing relaxed myosin states in hypertrophic cardiomyopathy by second harmonic-generation microscopy

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eLife Assessment

Arecchi et al. demonstrate that polarized second-harmonic generation microscopy can be used to probe the ON/OFF states of myosin in both permeabilized and intact muscle, making this key measurement accessible to a greater number of labs. This has the potential to help with the study of disease-causing mutations and our understanding of drug function. The methodology is well defined, and the results are **important**; however, whilst this is overall a **convincing** study, there are some limitations to the interpretation of the data.

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Abstract

This study explores the use of polarized second-harmonic generation (pSHG) to investigate myosin conformation in the relaxed state, differentiating between the actin-available, disordered (ON) state and the energy-conserving, ordered (OFF) state. By pharmacologically shifting the ON/OFF equilibrium using a myosin activator (2-deoxyATP) or inhibitor (Mavacamten), we demonstrate the sensitivity of pSHG in quantifying the ON/OFF ratio in skeletal and cardiac tissues. Comparisons with X-ray diffraction measurements further validate our findings. Applying this approach to a sarcomeric mutation associated with hypertrophic cardiomyopathy, we show that R403Q/MYH7-mutated minipigs exhibit a higher ON fraction compared to controls. This difference is abolished under high concentrations of myosin modulators, indicating saturation toward either state. ATPase assays reveal increased resting ATPase activity in R403Q samples, persisting even when the ON state is saturated, suggesting that elevated energy consumption in the R403Q mutation is driven by both an ON shift and enhanced myosin ATPase activity.

Introduction

Exploring the molecular structural dynamics within living cells that drive biological functions poses significant challenges due to the wide range of spatial and temporal scales involved. A classic example is the collective chemomechanical action of myosin motors within the sarcomere, the fundamental structural unit of striated muscle. Contractile forces in muscle are generated through the “power stroke,” a series of structural transitions in the actomyosin cross-bridge driven by ATP hydrolysis. Techniques such as cryo-electron microscopy (Cryo-EM) (1), electron paramagnetic resonance (EPR) (2), X-ray diffraction (3), fluorescence polarization (4), and, more recently, second-harmonic generation (SHG) (5) have provided detailed insights into the conformational changes of myosin motors during the power stroke.

Among these, SHG - a nonlinear optical process occurring in hyperpolarizable materials, including muscle - has proven particularly informative (6). In sarcomeres, SHG signals display a distinct pattern of alternating bright and dark bands that reflect the underlying striated architecture. This label-free method enables high-resolution probing of sarcomeres in intact tissues, making it an invaluable tool for studying sarcomere organization (7, 8). However, the potential of SHG extends beyond structural imaging. By varying the polarization of the incident light in single muscle fibres, it is possible to gain insights into structural states of actomyosin motors myosin conformations showing significant responsiveness to myosin conformational changes during muscle contraction (9, 10). Here, we further expand the application of SHG polarization anisotropy (pSHG) to investigate whether it can detect the distribution of conformations of myosin heads in relaxed striated muscle.

Emerging evidence suggests that, in relaxed muscle (i.e. when cytoplasmic calcium concentration falls below the threshold of thin filament activation) a fraction of myosin motors stick out of from the thick filament backbone while others are folded onto the thick filament surface. Folded heads assume a compact conformation or “interacting head motif structure” (11–15) shown by the reconstruction of the vertebrate thick filament of striated muscle (16, 17). The fraction of heads folded onto thick filament has been correlated to a “sequestered” or “OFF” state that could be the structural counterpart of the functional “super-relaxed state” (18, 19) characterized by reduced (10 times) ATPase turnover compared to detached myosin heads in the “disordered relaxed” or “ON” state. Within sarcomeres, upon calcium activation, the fraction of myosin heads in the ON state participates in initial force production, while the remaining heads, folded on the filament, represent a “reserve” of energy-conserving motors. A thick filament regulatory mechanism has been proposed to regulate the release of myosin motors from a folded conformation, complementing the classic calcium-induced thin-filament mechanism (20), especially in the modulation of heart muscle contraction (21, 22). Biochemical and structural studies have shown that the ratio between OFF and ON heads is significantly altered in congenital cardiovascular diseases such as hypertrophic cardiomyopathy (HCM) and heart failure (23), leading to mechanical and energetic dysfunctions (24, 25). Understanding myosin conformational states in the relaxed state is therefore critical for elucidating their role in force generation and energy regulation.

In this study, we used pSHG to quantify imbalances in myosin head conformations in both skeletal and cardiac muscle, including cardiac samples carrying the R403Q-MYH7 mutation, a well-established cause of HCM. (26, 27). We examined how this mutation affects myosin conformation by pharmacologically modulating the ON-OFF balance with activators (e.g., 2-deoxyATP (28, 29)) and inhibitors (e.g., mavacamten (30, 31)). These findings were directly compared with complementary structural data obtained through X-ray diffraction. Additionally, ATPase activity assays were performed to correlate structural alterations with changes in energy consumption. By integrating structural and functional analyses of ON-OFF

dynamics, this work aims to elucidate the role of the R403Q mutation in the hypercontractility phenotype of HCM and to provide new insights into the energy-regulating mechanisms of myosin in cardiac muscle.

Materials and methods

Second-harmonic generation microscope and pSHG imaging

Experimental setup

A pulsed titanium -sapphire laser (Chameleon Ultra II, Coherent), operating at 740 nm with a pulse width of 140 fs and a repetition rate of 80 MHz, was used for excitation. A laser scanner (Rapid Multi Region Scanner, MBF Bioscience, Vidrio Technologies) equipped with two galvanometric mirrors and a resonant mirror was employed to scan the excitation beam across the sample. To vary the polarization of the excitation laser, we used a quarter-wave plate (WPQ10E-780, Thorlabs) to achieve a well-defined circular polarization, followed by a rotating polarizer (ELL14K and LPNIRB100-B, Thorlabs) to generate uniform linear polarization with limited intensity variation ($I_{pp} < 1\%$ over 2π). Polarized light was focused on the sample using a 40× immersion objective (W-Plan Apochromat, 40×, 1.0 NA; Zeiss), while the SHG signal was collected in the forward direction using a 40× water immersion objective (Achromplan, 40×, 0.8 NA; Zeiss). A narrow band-pass filter (390/10 nm; FBH390-10, Thorlabs) placed before a GaAsP photomultiplier module (H11706P-40; Hamamatsu) was used to selectively detect the SHG signal.

SHG images of muscle tissue were acquired without any exogenous labelling. To characterize the polarization dependence of the SHG signal (pSHG) in the muscle tissue, SHG images were collected with different polarization angles (ϕ) of the incident light relative to the fibre axis. The entire field of view (FOV) was first examined to assess sample quality and identify regions of interest (ROIs) where sarcomeres exhibited minimal structural disarray with a sarcomere length of $2.1 \mu\text{m}$. Once well-organized regions were identified, ROIs of $14 \times 14 \mu\text{m}$ were imaged at 50 frames per second while the polarization angle rotated at 45 rpm. The excitation laser power (measured at the sample) was set to 30 mW. For each SHG image, the SHG intensity was integrated over all pixels, producing a polarization anisotropy intensity profile (I_{SHG}) as a function of the polarization angles (ϕ). Experimental data were then fitted with [Equation 1](#), which describes the theoretical polarization modulation of I_{SHG} under the assumption of cylindrical symmetry ([9](#)):

$$I_{\text{SHG}}(\phi) \propto (\sin^2(\phi) + \gamma \cos^2(\phi))^2 + \sin^2(2\phi) \quad [1]$$

This fitting procedure allows extraction of the geometrical parameter γ , which is linked to the overall geometrical organization of SHG emitters within the excitation volume. The fitting procedure was carried out using a custom-made LabVIEW program (National Instruments, version 12.0f3 at 32 bit).

Myofibril and skinned muscle strips

Rabbits were euthanized by pentobarbital administration (120 mg/kg) through the marginal ear vein, according to the procedure established by the European Union Council on the Use of Laboratory Animals (Directive 2010/63/EU), and using protocols approved by the Ethics Committee for Animal Experiments of the University of Florence. After careful dissection, the psoas muscle was cut into strips ~ 0.5 cm wide, and samples were tied at rest length to rigid wood sticks, and stored at -20°C for no more than 6 months in a 200 mM ionic strength rigor solution (100 mM KCl, 2 mM MgCl_2 , 1 mM EGTA, 50 mM Tris, pH 7.0) supplemented with glycerol 50%. Single myofibrils or bundles of two to three myofibrils were prepared by homogenization of glycerinated psoas muscle as previously described ([32](#)).

Skinned muscle strips were obtained from rabbit psoas glycerinated muscle strips, from 6 frozen minipig hearts, and from 6 mouse hearts. We employed a knock-in minipig model carrying the R403Q mutation in MYH7, generated by introducing the mutation together with a BlastR selection cassette, which was subsequently excised via Cre-mediated recombination (31). The successful integration of the mutation was confirmed by Southern blot analysis using probes specific for MYH7 and BlastR, as well as by PCR with primers flanking the BlastR cassette, demonstrating its effective removal following Cre treatment. Isolated hearts from 1-month-old WT and R403Q male Yucatan pigs were provided by Exemplar Genetics Inc. Humane euthanasia and tissue collection procedures were approved by the Institutional Animal Care and Use Committees at Exemplar Genetics. The procedures were conducted according to the principles in the “Guide for the Care and Use of Laboratory Animals” (33), the Animal Welfare Act as amended, and with accepted American Veterinary Medical Association (AVMA) guidelines at the time of the experiments (34). Frozen left ventricular tissues were obtained from R403Q and WT minipig hearts and stored at -80 °C until the experiments. Each R403Q minipig was a sibling of a WT minipig.

C57BL/6J mice from Envigo (N=6, 6-7 months old) were used in this study. All animal handling and procedures were conducted in accordance with Directive 2010/63/EU of the European Parliament on the protection of animals used for scientific purposes. The experimental protocol was approved by the Italian Ministry of Health (reference code: ODD9B.N.DZR). Mice were euthanized via inhaled isoflurane overdose (5%). Death was confirmed by the absence of a heartbeat, assessed through palpation, and the lack of thoracic movements. Once confirmed, the heart was explanted. Excised hearts were transported to the isolation station in a beaker containing 0.2 ml of heparin (5000 IU/ml) to prevent blood clotting, along with 10 ml of Krebs-Henseleit (KH) solution. Upon arrival at the isolation station, the hearts were immediately cannulated through the proximal aorta and retrogradely perfused with a KH solution modified to prevent spontaneous cardiac activity and to minimize contractures caused by shear damage during the dissection and placed in a dissecting chamber under a binocular microscope. KH contained (mM): 120 NaCl, 5 KCl, 2 MgSO₄, 1.2 NaH₂PO₄, 20 NaHCO₃, 0.50 Ca²⁺ and 10 glucose, pH 7.38 equilibrated with HEPES buffer. During dissection, the potassium concentration was adjusted to 15 mM to halt spontaneous heartbeats. Additionally, 2,3-butanedione monoxime (BDM) at 20 mM was added to prevent contractions during surgery. The solution was continuously oxygenated with O₂.

The ventricular muscle strips were incubated overnight in a cold (4° C) standard relaxing solution added with 1% Triton X-100, to remove cellular membranes and all cytoplasmic organelles membranes, such as mitochondria and sarcoplasmic reticula. After this skinning procedure, the tissue was washed using a fresh relaxing solution containing protease inhibitors (AP). Under a stereoscope, the de-membranated strips were cut into smaller pieces, following the alignment of the fibres. Selected skinned muscle strips were attached to custom-made aluminium foil T clips and mounted horizontally between a force transducer and a length control motor in the experimental chamber. The length of the preparations was adjusted to a sarcomere length of $\approx 2.20 \mu\text{m}$. The dimensions of the skinned strips were determined using an ocular micrometer mounted on the experimental microscope: on average, the preparations were 1 mm long, 140 μm wide, and 150 μm thick.

Myofibril and skinned preparations were pre-incubated in a series of relaxing solutions as follows: Relax (5 minutes), 2-deoxyATP (15 minutes), and Mavacamten (30 minutes), with imaging conducted after each incubation. Relax solutions: Relaxing solutions were calculated as described previously at pH 7.0, and contained 10 mM of total EGTA, 5 mM MgATP, 1 mM free Mg²⁺, 10 mM 3-(N-morpholino) propane sulfonic acid, propionate and sulfate to adjust the final solution to an ionic strength of 200 mM and a monovalent cation concentration of 155 mM. Phospho Creatine (10 mM) and Creatine Phospho Kinase (200 U ml⁻¹) were added to all solutions to minimize alterations in the concentration of MgATP and its hydrolysis products. [Pi] resulting from contaminant inorganic phosphate was less than 150 μM . Different pCa > 8.00 with 10 mM EGTA were used for the experimental protocol:

- pCa > 8.00, 5 mM ATP
- pCa > 8.00, 5 mM 2-deoxyATP (Cat # C01577; Genscript, USA)
- pCa > 8.00, 5 mM ATP, and Mavacamten (Axon medchem BV -Groningen, The Netherlands; 50 μ M in psoas muscle and 10 μ M in cardiac preparation)

All solutions to which the samples were exposed contained a cocktail of protease inhibitors including leupeptin (10 μ M), pepstatin (5 μ M), phenylmethylsulphonyl fluoride (200 μ M), and E-64 (10 μ M), as well as NaN₃ (500 μ M) and 500 μ M dithiothreitol.

For imaging in rigor an 200 mM ionic strength solution was used, containing 100 mM KCl, 2 mM MgCl₂, 1 mM EGTA, 50 mM Tris adjusted to pH 7.0

Intact mouse cardiac preparations

Unbranched trabeculae extending from the free ventricular internal wall to the atrioventricular fibrous skeleton were dissected from mouse hearts by cutting their attachments: one end at the atrioventricular valve and the other at the ventricular wall. Trabeculae, with diameters ranging from 150 to 400 μ m, were mounted between the hook-shaped extensor of two micromanipulators and placed in the imaging chamber equipped with two platinum electrodes for electrical stimulation. During imaging, the mouse trabeculae were stimulated at 0.1 Hz and continuously oxygenated. They were initially pre-incubated in KH solution for 15 minutes before the first imaging session. Subsequently, the preparations were exposed to KH containing 10 μ M Mavacamten and incubated for 30 minutes prior to the second imaging session. All chemicals and enzymes, except for 2-deoxyATP and Mavacamten, were purchased from Sigma-Aldrich (Merck Life Science, St. Louis, MO, USA).

X-ray diffraction measurements

Experimental setup

X-ray diffraction experiments were performed using the small angle X-ray diffraction instrument on the BioCAT beamline 18ID at the Advanced Photon Source, Argonne National Laboratory (35). The X-ray beam energy was set to 12KeV (0.1033 nm wavelength) at an incident flux of $\sim 5 \times 10^{12}$ photons per second. The X-ray beam was focused to $\sim 250 \mu\text{m} \times 250 \mu\text{m}$ (horizontal and vertical respectively) at the sample position and to $\sim 150 \times 30 \mu\text{m}$ at the detector. The specimen to detector distance was about 3.5 m. The muscles were incubated in a customized chamber and experiments were performed at 28-30 °C. The muscles were stretched to a sarcomere length of 2.3 μm by monitoring the laser diffraction patterns from a helium-neon laser (633 nm). The X-ray patterns from either WT or R403Q samples were collected either in relaxing solutions with 100% ATP and then 100% 2-deoxyATP or in relaxing solution and then in relaxing solution with 50 μ M Mavacamten on a MarCCD 165 detector (Rayonix Inc., Evanston IL) with a 1 s exposure time. The muscle samples were oscillated along their horizontal axes at a velocity of 1 - 2 mm/s to minimize radiation damage. The irradiated areas are moved vertically after each exposure to avoid overlapping X-ray exposures. 2-3 patterns were collected under each condition. The data were analysed using the MuscleX software package (36). The X-ray patterns were quadrant folded and background subtracted using the “Quadrant Folding” routine in MuscleX. The intensities and spacings of meridional and layer line reflections were measured by the “Projection Traces” routine in MuscleX as described previously (37). To compare the intensities under different conditions, and correct for any possible X-ray diffraction intensity changes due to radiation damage to the muscle, the measured intensities of X-ray reflections are normalized to the total diffuse background intensity as described previously (38). There were no significant changes in

the radial width of the M3 reflection among different conditions, thus no correction for width was applied to the measured intensities. Values obtained from patterns under the same condition were averaged.

Sample preparation and solutions

Previously frozen WT and R4303Q left ventricular myocardium samples from 1-month old Yucatan mini-pig were prepared as described previously (39, 40). Briefly, samples were defrosted in skinning solution (2.25 mM Na₂ATP, 3.56 mM MgCl₂, 7 mM EGTA, 15 mM sodium Phospho Creatine, 91.2 mM potassium propionate, 20 mM Imidazole, 0.165 mM CaCl₂, Creatine Phospho Kinase 15 U/ml with 15 mM 2,3-Butanedione 2-monoxime (BDM) and 1% Triton-X100 and at pH 7) at room temperature for 30 min before splitting to smaller bundles (500 µm in diameter and variable lengths). The fibre bundles were further skinned in fresh skinning solution at room temperature for 2-3h. Muscles were then washed with fresh relaxing solution (2.25 mM Na₂ATP, 3.56 mM MgCl₂, 7 mM EGTA, 15 mM sodium Phospho Creatine, 91.2 mM potassium propionate, 20 mM Imidazole, 0.165 mM CaCl₂ Creatine Phospho Kinase 15 U/ml at pH 7) for ~10 minutes, repeated 3 times. The tissue was dissected into ~200-300 µm diameter, 3-5 mm long fibre bundles and clipped with aluminium T-clips. The preparations were then stored in cold (4°C) relaxing solution with 3% dextran for the day's experiments. The muscles were incubated in a customized chamber, and experiments were performed at 28-30 °C.

ATPase measurements

Experimental setup

The experimental apparatus and the principle of the ATPase measurement were described in (41, 42). Briefly, the mounted muscle preparations were manually transferred between different baths, to expose them to different relaxing (pCa > 8.00) solutions: Relaxing, 2-deoxyATP Relaxing solution, and Mavacamten Relaxing solution, which were used subsequently for the measurements. The bath system consisted of an 80 µL well for incubating samples in relaxing solutions and a second well dedicated to the ATPase assay. This assay bath, with a volume of 30 µL, featured quartz windows to allow the transmission of near-UV light (340 nm) for measuring NADH absorbance. The bath was continuously stirred by motor-driven vibration of a membrane positioned at its bottom. The baths were milled in aluminium blocks (anodized) and mounted on top of an aluminium base, through which water was circulated to allow temperature control of all solutions (21 °C). ATPase activity measurements were normalized to the volume: length × cross-sectional area (CSA) of the muscle strip. The CSA was estimated in the setup assuming an elliptical shape: (width × depth × π)/4.

Solutions

Different Relaxing solutions of pCa > 8.00 with 10 mM EGTA were used for the experimental protocol:

- Control Relaxing: pCa > 8.00, 6.2 mM ATP
- 2-deoxyATP Relaxing: pCa > 8.00, 6.2 mM 2-deoxyATP
- Mavacamten Relaxing: pCa > 8.00, 6.2 mM ATP, and Mavacamten 10 µM

All solutions contained: 100 mM BES (N,N-bis[2-hydroxyethyl]-2-aminoethane sulphonic acid); 6.63 mM MgCl. In addition, all solutions contained 0.25 mM NADH, 0.5 mg/mL pyruvate kinase (500 U/mg), 0.1 mg/mL LDH (912 U/mg), 10 µM oligomycin B. Oligomycine together with sodium azide 5 mM were added to suppress any residual mitochondrial ATPase or ATP synthase activity. The ionic strength of the solutions was maintained at 200 mM by adding the appropriate amount of

potassium propionate (KProp). The pH was adjusted to 7.00 with KOH. Aliquots of these solutions (minus NADH) were stored frozen. On the day of the experiment, the solutions were thawed and the solutions were kept on ice until required.

All chemicals were purchased from Merck life Science

Results

pSHG sensitivity to myosin conformation in psoas skeletal muscle

pSHG microscopy ([Fig. 1A](#)) was first applied to psoas skeletal muscle, which has a well-organized structure, to investigate whether it can detect myosin conformations during the relaxation state. SHG images are characterized by a remarkable alternation of light and dark bands, reflecting the sarcomeric striations typical of the muscular tissue ([Fig. 1B](#)). pSHG recordings were performed in selected ROIs in rigor, in relax solution, and treated with a high concentration (50 μ M) of the myosin motor inhibitor Mavacamten. The three experimental conditions highlight different pSHG intensity modulations, which can be well described by the theoretical polarization modulation of I_{SHG} ([Eq. 1](#)), resulting in distinct γ values ([Fig. 1C](#)). Repeated measurements on different rigor fibres yielded an average value of $\gamma_{\text{Rigor}} = 0.68 \pm 0.01$ (mean $\pm$ SEM) while the same measurement performed on demembranated relaxed fibres produced an average value of $\gamma_{\text{Relax}} = 0.39 \pm 0.01$ ([Fig. 1D](#)). The large statistically significant difference between γ_{Rigor} and γ_{Relax} confirms the capability of pSHG to probe distinct conformation changes of myosin motors. Notably, by applying Mavacamten we observed a further reduction in the γ value, which decreased significantly from the baseline relaxation state, with a $\gamma_{\text{Mava}} = 0.27 \pm 0.01$.

The reduction of γ in the presence of Mavacamten is consistent with a conformation change of the myosin in approaching the thick filament backbone ([30](#)). The clear shift in the γ value corroborates the expected effects of Mavacamten and demonstrates the efficacy of our technique in probing myosin conformational states under relaxing conditions. A similar behaviour was observed in isolated myofibrils, where gamma values were comparable to those measured in skinned preparations (see [Fig. S1](#)).

pSHG sensitivity in demembranated and intact cardiac preparations

We further evaluated the pSHG sensitivity to distinct relaxed myosin states by applying the same experimental approach to mouse ventricular preparations ([Fig. 2A](#)). The cardiac SHG images also displayed regular alternation of bright and dark bands, visually underscoring the uniformity of the sarcomere architecture. We started with measures of skinned mouse cardiac samples in rigor, relaxed, and with a high concentration of Mavacamten (10 μ M). Consistent with our observations in the psoas muscle, we observed a significant difference of γ value between rigor and relaxed and a further reduction in relax state after applying Mavacamten ([Fig. 2B](#)).

We also examined whether γ values were reflected in intact cardiac preparations. To this end, we conducted pSHG measurements in murine isolated trabeculae stimulated at 0.1 Hz, allowing us a diastolic phase duration compatible with the pSHG scan. As with skinned muscle preparations, we observed a statistically significant decrease in γ values both in skinned and in intact preparations after the pharmacological treatment ([Fig. 2C](#)). The γ values in intact samples were consistent with those observed in demembranated strips, both in the relaxing solution and after the addition of Mavacamten. These data demonstrated the effectiveness of pSHG in probing distinct structural conformations of myosin *in vivo* at rest, reinforcing the robustness and reliability of the method across different experimental conditions.

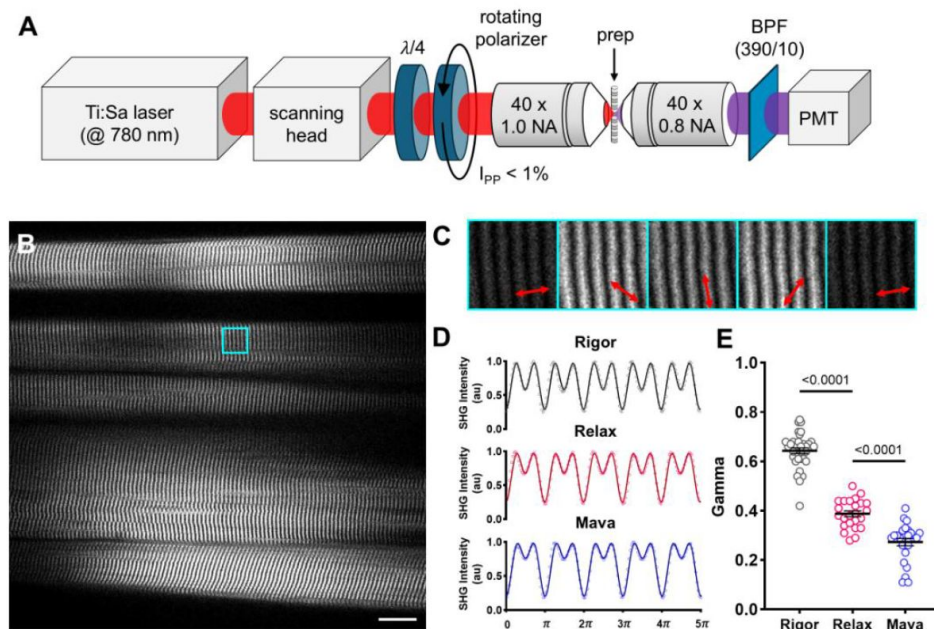


Figure 1

pSHG sensitivity on myosin conformation in psoas skeletal muscle.

A) Optical scheme of the SHG microscopy setup. The excitation beam's optical path is shown in red, while the emission path is in purple. Abbreviations: $\lambda/4$, quarter-wave plate; BPF, bandpass filter; PMT, photomultiplier. B) SHG image of a demembrated rabbit psoas fibre. Bright bands correspond to sarcomeric A bands. Scale bar: 20 μm . The cyan square highlights a representative region of interest (ROI) selected for acquisition ($13 \times 13 \mu\text{m}$). C) Selected ROI shown in panel B illuminated at different polarization angle. D) Representative SHG polarization curves in rigor, relaxation, and after exposure to a high concentration of Mavacamten (Mava; 50 μM). Circles represent raw data, and the continuous lines indicate the best fit of Eq. 1, yielding a best-fit parameter of $\gamma_{\text{Rigor}} = 0.68$ in rigor, $\gamma_{\text{Relax}} = 0.36$ in relax, and $\gamma_{\text{Mava}} = 0.27$ with Mavacamten. E) Graph illustrating γ values among the Rigor state, Relax state, and after exposure of Mavacamten (Mava) in rabbit skinned psoas fibers. Data were obtained from 6 psoas fibres, with multiple ROIs collected across each strip, producing 35 data points for the rigor state, 24 for relaxation, and 26 for Mavacamten. Data are reported as mean $\pm$ S.E.M. A one-way repeated measures ANOVA was performed with a Tukey post-hoc correction.

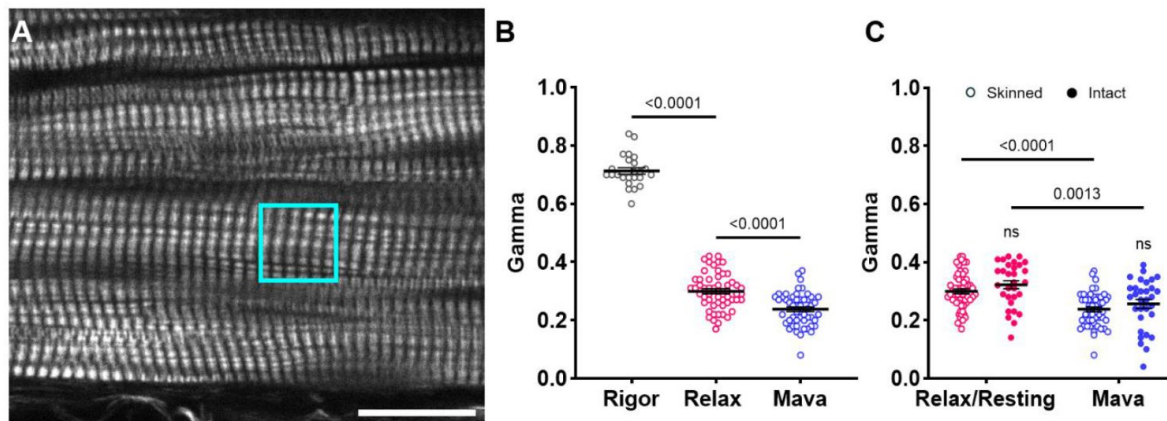


Figure 2

pSHG sensitivity on demembranated and intact cardiac preparations.

A) Representative SHG image of a demembranated mouse ventricular wall strip. Borders between individual cells are visible. Scale bar: 20 μm . The cyan square highlights a representative region of interest (ROI) selected for acquisition (13 $\times$ 13 μm). B) Graph illustrating γ values among the Rigor state, Relax state, and after exposure of Mavacamten (Mava) in mouse skinned cardiac samples. Data were obtained from 5 skinned cardiac samples dissected from 3 mice. Multiple ROIs were collected across each strip, producing 18 data points for the rigor state, 18 for relaxation, and 18 for Mavacamten. Data are reported as mean $\pm$ S.E.M. A one-way repeated measures ANOVA was performed with a Tukey post-hoc correction. C) Comparison of γ values in mouse demembranated versus intact cardiac preparations. The graph shows γ values measured in both skinned (open circles) and intact (filled circles) multicellular cardiac preparations. Data were collected first in a relax solution and subsequently after the addition of 10 μM Mavacamten. For intact preparations, stimulation was applied at a frequency of 0.1 Hz, with imaging performed during the long diastolic phase. Data were obtained from 5 skinned cardiac samples dissected from 3 mice and 6 intact trabeculae dissected from 3 mice. Multiple ROIs were collected across each strip, producing 55 data points for the relax state and 52 for Mavacamten in skinned preparation while 30 data points for the resting state and 30 for Mavacamten were obtained from intact preparations. Data are reported as mean $\pm$ S.E.M. A two-way repeated measures ANOVA was performed with a Tukey post-hoc correction.

Probing ON state in R403Q-MYH7 mutation

Driven by these results, we employed pSHG to quantify the ratio between the disordered-relaxed (ON) and sequestered (OFF) states in a minipig model carrying an HCM-associated mutation (R403Q-MYH7). Myocardial disarray is one of the structural abnormalities classically associated with HCM, and its presence contributes to contractile dysfunction and energetic impairment (43 [↗](#)). To minimize polarization artefacts potentially caused by structural disarray - particularly evident in pathological preparations (**Fig. 3A** [↗](#)), the entire sample was initially examined to carefully identify ROIs where sarcomeres displayed minimal disarray.

Data acquisition was collected in these well-organized regions. From pSHG measurements in the relaxed condition (**Fig. 3B** [↗](#)), we observed a γ value in R403Q-MYH7 that was significantly higher than that measured in sibling wild-type minipigs (WT). Notably, the higher γ value in R403Q-MYH7 is consistent with an increased level of ON state myosin heads associated with this mutation.

We further carried out pharmacological manipulation to compare the effects of 2-deoxyATP, a naturally occurring ATP analog that binds to myosin catalytic site promoting a shift toward the ON state- and Mavacamten, which drives the equilibrium toward the OFF state, in preparations expressing the R403Q mutation versus WT. As expected, the application of 2-deoxyATP or Mavacamten resulted in an increase or reduction in γ values, respectively, in both preparations. Of note, the increase in the γ value observed with 2-deoxyATP is not linked to actomyosin complex formation, as no detectable increase in passive force was observed in the presence of 2-deoxyATP (see Fig. S2).

In the presence of either of these two molecules the original differences between WT and R403Q groups disappeared, suggesting that we had effectively saturated the disordered state with 2-deoxyATP and the OFF state with Mavacamten. This observation is particularly intriguing as it allows us to define an upper limit (2-deoxyATP) and a lower limit (Mavacamten) for the ON population, enabling the quantification of the fraction of ON myosin heads within the tested tissue. Under relaxed conditions, we observed a notable increase in the proportion of disordered myosin heads, rising from $\approx 30\%$ in WT samples to $\approx 65\%$ in R403Q mutant preparations (**Fig. 3C** [↗](#)).

To enhance the robustness of our findings, complementary X-ray diffraction measurements were performed (**Fig. 4** [↗](#)). X-ray diffraction patterns obtained from R403Q minipig model in a relaxing solution show a marked decrease in the intensity of the myosin-based M3 meridional reflection compared to the WT littermates. Among myosin-based reflections, the M3 meridional reflection originates from the axial repeat of myosin motors, indicating an increase in disordered heads in R403Q. Consistently, the spacing of the M6 meridional reflection (SM6), originating from a periodic mass distribution in the thick filament backbone, is enhanced in R403Q compared to WT. As expected, upon the application of 2-deoxyATP and Mavacamten on the same strips (paired measurements), we observed that R403Q preparations, compared to WT, there was reduced sensitivity to 2-deoxyATP but increased sensitivity to Mavacamten. These findings further support that the R403Q mutation shifts myosin towards the ON state, underlying the hypercontractility characteristic of R403Q-induced HCM.

ATPase activity measurements in R403Q mutant myocardium

To provide a more comprehensive understanding of the implications of the R403Q mutation, alongside the pSHG analysis, we performed ATPase activity measurements to evaluate the specific effects of the R403Q mutation on sarcomere energetics at rest. This correlative approach enabled us to link structural observations with functional assessments, deepening our understanding of the R403Q mutation's influence on the ON and OFF states of myosin.

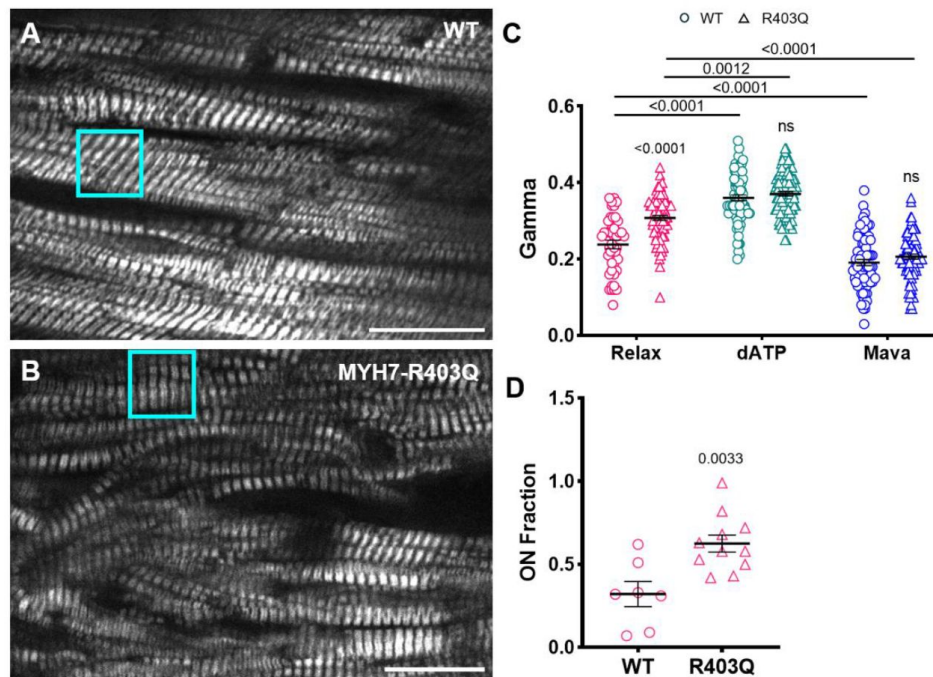


Figure 3

pSHG in R403Q-MYH7 mutation

A) SHG image of a sample from a wild-type (WT) minipig, and B) sample from a minipig carrying the MYH7-R403Q hypertrophic mutation. Cyan squares highlight representative regions of interest (ROIs) selected for acquisition ($13 \times 13 \mu\text{m}$). A marked difference is evident between healthy and pathological tissue, with the mutated tissue exhibiting a high degree of structural disarray. Scale bars: $20 \mu\text{m}$. C) Pharmacological manipulation of skinned preparations from WT and R403Q minipigs. The graph shows γ values obtained under different conditions: in Relax solution, after exposure to 100% 2-deoxyATP (dATP), and following treatment with $10 \mu\text{M}$ Mavacamten (Mava). Data were obtained from 7 skinned cardiac samples dissected from 3 WT minipig and 11 skinned cardiac samples dissected from 3 R403Q-MYH7 minipig. Multiple ROIs were collected across each strip, producing 45 and 76 data points for the relax state, 77 and 103 for 2-deoxyATP, and 76 and 91 for Mavacamten in WT and R403Q-MYH7, respectively. Data are reported as mean $\pm$ S.E.M. A two-way repeated measures ANOVA was performed with a Tukey post-hoc correction. D) Estimation of the fraction of ON state myosin heads in WT and R403Q preparations. This analysis assumes complete saturation of ON and OFF states following treatment with 100% 2-deoxyATP and $10 \mu\text{M}$ Mavacamten respectively. Unpaired T-test analysis was performed.

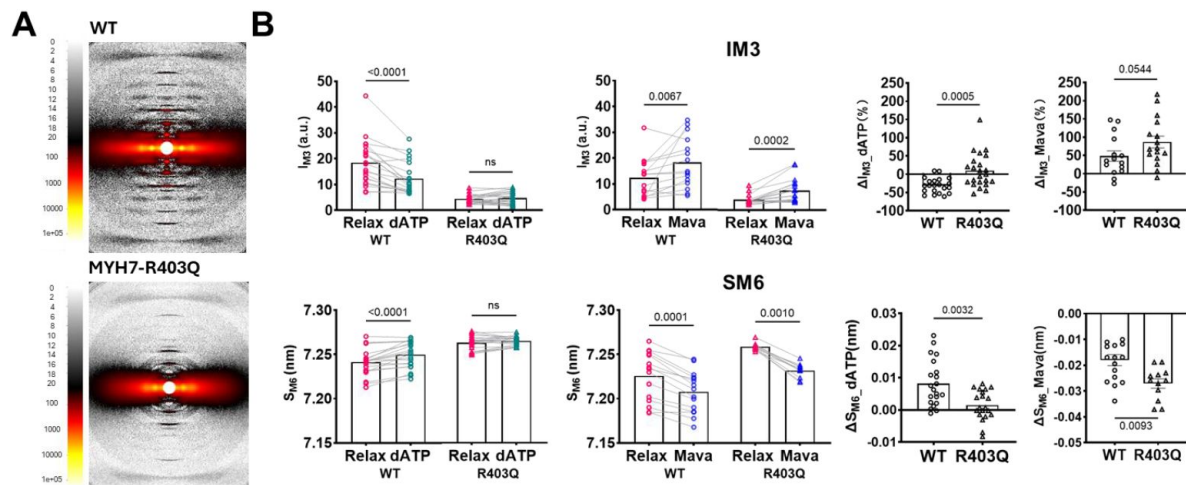


Figure 4

X-ray diffraction in R403Q-MYH7 mutation.

A) Representative X-ray diffraction patterns from wild-type (WT) and R403Q minipig ventricular strips in relaxing solution. B) Top row: Graphs comparing changes in M3 reflection intensity (IM3) between WT and R403Q samples under different conditions. The first two graphs show intensity differences between Relaxed and 2-deoxyATP (dATP) treated samples (WT and R403Q, respectively) and between Relaxed and Mavacamten (Mava) treated samples. The third and fourth graphs illustrate ΔIM3 , comparing the changes in IM3 between WT and R403Q samples with 2-deoxyATP and Mavacamten treatments. Bottom row: Graphs comparing the spacing of the M6 meridional reflection (SM6) between WT and R403Q samples under different conditions. The first two graphs display shifts in SM6 between Relaxed and 2-deoxyATP treated samples and between Relaxed and Mavacamten treated samples for both groups. The third and fourth graphs show ΔSM6 , comparing shifts in SM6 between WT and R403Q with 2-deoxyATP and Mavacamten treatments. Data were obtained from > 15 skinned cardiac samples dissected from 2 WT minipig and 12 skinned cardiac samples dissected from 2 R403Q-MYH7 minipigs.

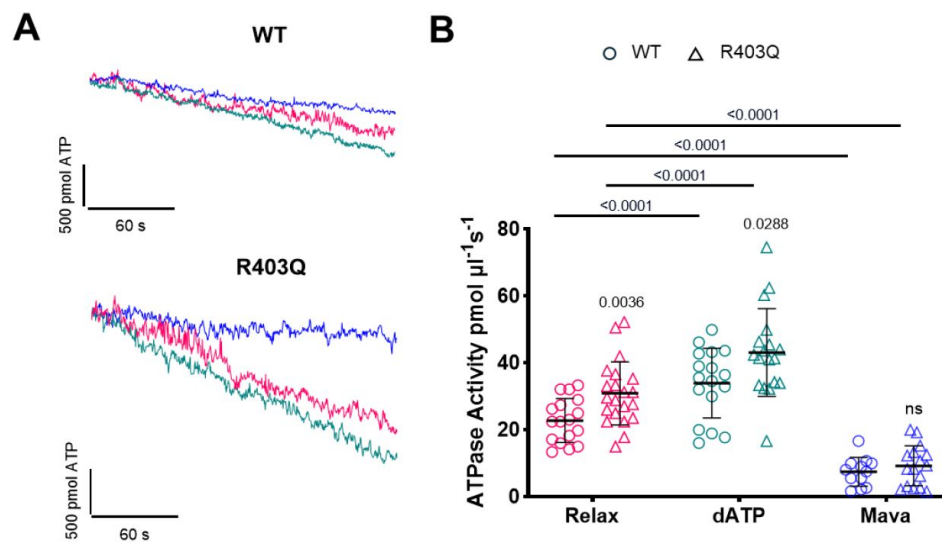


Figure 5

ATPase activity in R403Q mutation.

A) Representative traces of NADH absorbance from wild-type (WT) minipig preparations (left column) and preparations from minipigs carrying the MYH7-R403Q hypertrophic mutation (right column) under different conditions: Relax, 100% 2-deoxyATP (dATP), and 10 μM Mavacamten (Mava). B) Graph illustrating ATP consumption across different sample conditions: in Relaxing solution, after exposure to 100% 2-deoxyATP, and following treatment with 10 μM Mavacamten. Data were obtained from 17 skinned cardiac samples dissected from 3 WT minipig and 21 skinned cardiac samples dissected from 3 R403Q-MYH7 minipig. Each preparation is associated with a data point in the three conditions. Data are reported as mean $\pm$ S.E.M. A mixed effects ANOVA was performed with a Tukey post-hoc correction.

As expected, a significant difference in ATPase activity was observed between WT and R403Q preparations, with the latter exhibiting markedly higher energy consumption compared to the WT counterparts (**Figure 4**). This finding aligns with the structural analysis, where mutant preparations showed a shift toward the ON state. Exposure to a high concentration of Mavacamten (10 μ M) resulted in a global reduction in energy consumption in both preparations, bringing ATPase activity to comparable levels. These results are consistent with the structural analysis, which showed a complete shift of myosin heads toward the OFF state in both WT and R403Q samples. On the other hand, in the presence of 2-deoxyATP, which by itself is associated with increased ATPase activity, a significant disparity in energy consumption between the two groups remained, with the R403Q mutation associated with a higher energy requirement. Under the same conditions, the structural data indicated that both WT and R403Q samples underwent a complete transition to the ON state, with no significant differences in their structural configurations.

This observation suggests that the increased energy consumption associated with the R403Q mutation is not solely explained by differences in the balance between the structural ON and OFF states in WT and R403Q myocardium.

Discussion

Thanks to the coherent summation of hyper-Rayleigh scattering from the amide groups (HN-CO) of polypeptide chains—which act as second-harmonic emitters—SHG signals can be endogenously detected in certain biological structures that possess the necessary spatial organization and symmetry. The organization of myosin into helical filaments and their arrangement into cylindrically symmetric, repetitive structures along the fibre provide an ideal configuration for generating SHG. In this context, the distribution of second-harmonic emitters can be probed by analysing the polarization modulation of SHG intensity (pSHG), which enables the derivation of a γ value associated with the emitters distribution angle relative to the principal axis of symmetry (fibre axis). However, to accurately correlate the γ value with the protein's structural conformation, it is essential to understand how the second-harmonic emitters are organized within the protein under investigation. By reconstructing each contributing second-harmonic emitter within the actomyosin motor array at the atomic scale, Nuccitelli et al. established a direct relationship between myosin conformation and the γ value. Notably, this revealed an overall trend: the γ value increases as the myosin angle relative to the thick filament backbone becomes larger, and decreases as this angle is reduced, opening the possibility of discerning between 'sequestered' (OFF state) or 'disordered' (ON state) myosin molecules in the relaxed state.

In this study, we employed pSHG to investigate the structural conformation of myosin in the relaxed state in skeletal (rabbit psoas) and cardiac (mouse and minipig) muscle by means of the γ value. Under relaxing conditions, pSHG analysis revealed a γ_{Relax} value of 0.39 ± 0.01 (mean $\pm$ SEM) in psoas skeletal muscle, 0.30 ± 0.01 in mouse cardiac muscle, and 0.25 ± 0.01 in minipig cardiac muscle. The difference in γ values between skeletal and cardiac muscle can be related to distinct structural arrangements of myosin heads in the relaxed thick filament, as well as variations in the basal ON/OFF state ratio. pSHG measurements under conditions that minimize ON state contributions (in the presence of high doses of Mavacamten) revealed an overall decrease in γ values. Specifically, γ values were reduced by approximately 30% in psoas skeletal muscle and 20% in cardiac preparations, suggesting that in our conditions (room temperature, no dextran) myosin heads are largely in an OFF conformation.

This is further supported by the resting ratio of ON/OFF heads quantified in WT minipig ($\approx 30\%$) cardiac samples, under the assumption that complete saturation of ON and OFF states is achieved following treatment with 100% 2-deoxyATP and 10 μ M Mavacamten, respectively. Although further validation is required to confirm that these pharmacological treatments achieve complete

saturation of the ON and OFF states, the estimated ON/OFF ratios are consistent with values obtained using X-ray diffraction and fluorescence-based methods under comparable experimental conditions in both skeletal (45) and cardiac muscle (46, 47). In contrast, the ON/OFF state ratio found here does not align with prior estimates obtained through mantATP biochemical assays (18, 31, 48–50). This discrepancy suggests that an overestimation of fast cycling heads (ON state) may occur even after correction for non-specific binding of the nucleotide. This contributes to the ongoing discussion in the field regarding the reliability of MantATP measurements (51–53), particularly in the context of the numerous unknown factors involved in correlating the structural and functional states of the myosin motor.

Interestingly, significant differences in γ values were found between skeletal and cardiac muscle even under pharmacological conditions that minimize the contribution of ON states (Fig. S3). This supports the hypothesis that differences in thick filament structure are associated with the prevalence of the different myosin isoforms. In skeletal muscle, particularly in the psoas, the predominant myosin heavy chain (MyHC) isoforms are IIx (MyHC1) and IIb (MyHC4), which are characteristic of type II muscle fibres, also known as fast-twitch fibres (54, 55). The dominant isoform in mouse cardiac ventricular muscle is alpha myosin heavy chain (MyHC6 or α -MyHC), typical of cardiac tissue in small mammals and atrial tissue in larger mammals (56, 57). Conversely, in minipig ventricular cardiac muscle, the main isoform is beta myosin heavy chain (MyHC7 or β -MyHC), characteristic of ventricular muscle in larger mammals, including humans (58, 59). The difference of γ values in demembranated muscle preparations of different origin in presence of Mavacamten then suggest a critical role of myosin isoforms in dictating the fine structure of the thick filament in the relaxed state.

pSHG was also employed to quantify the ON/OFF state ratio in preparations carrying the R403Q mutation. pSHG measurements show a clear increase in γ values in the R403Q mutation, consistent with the hypothesis of an increased ON population compared to WT. X-ray diffraction patterns obtained from the same R403Q minipig model show a marked decrease in the intensity of the third-order meridional reflection (M3) in the R403Q preparations compared to the WT littermates. In addition, the spacing of the sixth-order meridional reflection (SM6) increases significantly in the R403Q minipig preparations. The increase in the thick filament backbone periodicity, indicated by the increased SM6, and a reduction in the degree of helical ordering of the myosin heads, indicated by the reduced M3 intensity, are characteristic signatures of the structurally-defined OFF-ON transition of myosin (60). In line with this observation, we found that R403Q is less sensitive to 2-deoxyATP compared to WT, suggesting the R403Q mutation effects on OFF-ON transition are nearly as effective at 2-deoxyATP. Our additional observation of a pronounced effect of Mavacamten increase M3 intensity and SM6 spacing provides evidence that the R403Q mutation alters the conformation of myosin in the thick filament, favouring the ON state.

Furthermore, thanks to the sarcomere-level resolution provided by SHG imaging, pSHG ensures that γ values are not influenced by myofilament disarray observed in R403Q mutant samples. The differences observed during pSHG measurements primarily reflect variations in the ON/OFF state ratio between WT (30%) and R403Q (65%).

Taken together, these results support the widely accepted hypothesis that many HCM-associated β -cardiac myosin mutations destabilize the structural OFF states of thick filament-assembled myosin. This occurs by disrupting key interactions within the motor head and/or neck, leading to an increase in the number of heads available for interaction with actin and altering the ON/OFF ratio of motors participating in the cross-bridge cycle (25, 61–67). This view is consistent with the reported increase in resting energy consumption in the presence of R403Q β -cardiac myosin in cardiac sarcomeres (68, 69). An increase in the resting energy consumption associated with the R403Q mutation is also confirmed by our ATPase measurements, where we observe a higher ATP turnover in R403Q minipig cardiac strips compared to WT. Additionally, in

the presence of a high concentration of Mavacamten, ATPase measurements are significantly reduced in both R403Q and WT, resulting in statistically indistinguishable values. These data are in full agreement with the coupled pSHG measurements, which indicate that in the presence of Mavacamten, R403Q and WT are represented with a similar ON/OFF state fraction, with a complete transition to the OFF state. On the other hand, it is important to note that ATPase measurements carried out under 2-deoxyATP conditions show an additional significant increase in energy consumption for both WT and R403Q groups. It has been reported that 2-deoxyATP increases ATPase of acto-HMM (44 [44](#)) and myosin S1 (52 [52](#)), demonstrating the presence of increased basal ATPase activity independent of the ON/OFF configuration of myosin structure. Here we show that increased hydrolytic activity of myosin with 2-deoxyATP is also present despite changes in myosin structure induced by R403Q mutation. Data obtained from our correlative approach, which combines parallel structural and functional investigations, suggest that the R403Q mutation not only destabilizes the structural OFF states of thick filament-assembled myosin but also associates with an increase of intrinsic activity of myosin motors (68 [68](#)).

In addition, assuming that the presence of mavacamten reflects the activity of the OFF head and, conversely, the presence of 2-deoxyATP reflects the activity of the ON head, we can estimate the energetic ratio between the ON and OFF heads in both WT and R403Q. Assuming the absence of a non-myosin ATPase background and the same basal increase in ATPase induced by 2-deoxyATP, we found an ON/OFF energy ratio of approximately ≈ 3 and ≈ 4.5 for WT and R403Q, respectively. Although this value does not entirely match the one estimated using mantATP biochemical assays (≈ 10 (48 [48](#))), this discrepancy could be due to a non-specific ATPase background affecting our measurement and thus underestimating the energetic ratio.

The capability of pSHG to probe myosin dynamics structurally in physiological contexts, represents a novel complementary approach to well-established atomic-scale techniques like X-ray diffraction. This study also demonstrated the ability to monitor myosin conformational changes under relaxing conditions also *in vivo*, where pSHG showed γ values that were not statistically different from those observed in the same muscle in the skinned configuration. This finding not only opens the possibility of applying this method to intact multicellular preparations but also reinforces the reliability of the geometrical parameter γ as a tool for selectively probing myosin states. Although the micron-scale resolution of SHG microscopy inherently limits a comprehensive assessment of molecular conformation, the ability to probe the myosin state *in vivo* through polarization anisotropy without labelling, combined with its relatively low technological requirements, makes pSHG an accessible and valuable tool for any microscopy facility.

Future perspectives

Based on these findings, we propose that this technique could serve as a valuable tool for gaining deeper insights into the physiology and pathophysiology of muscle. For example, studying how temperature and lattice factors affect the ON/OFF ratio estimated by pSHG, in conjunction with ATPase measurement, could help to answer the still open question of how the structural IHM state correlates with the OFF or “SRX” state in resting muscle (53 [53](#)). In pathological contexts, numerous applications and studies could be envisioned. For example, the ON/OFF ratio at rest is thought to be primarily regulated by cardiac protein C (70 [70](#)), highlighting mutations in this protein—especially in cases of haploinsufficiency—as a critical area of research. The loss of protein C has been associated with an increased ON fraction, as demonstrated by other methodologies. Investigating mutations that induce similar shifts through entirely different mechanisms could provide valuable insights. Additionally, future studies might explore mutations in thin filaments, which appear to elicit comparable effects (71 [71](#)). This suggests that a variety of mutations may converge on a shared molecular ‘phenotype.’

Moreover, the temporal resolution of our pSHG approach (seconds) could enable the observation of myosin in stationary states, such as during the plateau phase of isometric contraction. To capture the rapid dynamics of the myosin working stroke, which occurs on a millisecond

timescale, the temporal resolution could be further enhanced by limiting pSHG measurements to two orthogonal angles—one parallel and one perpendicular to the fiber. This implementation could significantly improve the time resolution, expanding the applicability of this technique to the study of working stroke dynamics.

Lastly, the lab-scale setup of pSHG is well-suited for further implementation in high-throughput screening systems, where the ON/OFF ratio could be assessed across different pathological settings and patients in a multi-well configuration (72 [↗](#)).

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Additional files

Supplementary Figures [↗](#)

Additional information

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Reviewer #1 (Public review):

Summary:

This study utilizes polarized second-harmonic generation (pSHG) microscopy to investigate myosin conformation in the relaxed state, distinguishing between the disordered, actin-accessible ON state and the ordered, energy-conserving OFF state. By pharmacologically modulating the ON/OFF equilibrium with a myosin activator (2-deoxyATP) and inhibitor (Mavacamten), the authors demonstrate that pSHG can sensitively quantify the ON/OFF ratio in both skeletal and cardiac muscle. Validation with X-ray diffraction supports the accuracy of the method. Applying this approach to a hypertrophic cardiomyopathy model, the study shows that R403Q/MYH7-mutated minipigs exhibit an increased ON state fraction relative to controls. This difference is eliminated under saturating concentrations of myosin modulators, indicating that the ON/OFF balance can be pharmacologically shifted to its extremes. Additionally, ATPase assays reveal elevated resting ATPase activity in R403Q samples, which persists even when the ON state is saturated, suggesting that increased energy consumption in this mutation is driven by both a shift toward the ON state and inherently higher myosin ATPase activity.

Strengths:

This is a well-written and well-conducted study that clearly reveals the power of SHG microscopy. The study clearly establishes the great utility of SHG to study thick filament regulation.

Weaknesses:

- (1) Several studies have shown that the ON state of the thick filament is sensitive to both temperature and filament lattice spacing, with a common recommendation to conduct skinned fiber experiments at temperatures above 27°C and in the presence of dextran to better preserve physiological conditions. The authors should clarify the experimental temperature used in their skinned fiber studies, indicate whether dextran was included, and discuss whether adherence to these recommended conditions would have impacted their results.
- (2) On page 13, the authors report the proportion of disordered heads as approximately 30% in wild-type and 65% in R403Q fibers. They should clarify whether these values represent the percentage of total myosin heads, or rather the percentage of heads that are responsive to Mavacamten and dATP.
- (3) In Figure 5, regarding ATPase measurements, the content of contractile material per unit volume of muscle preparation will influence the results. Did the authors account for this variable, and if not, how might it have affected the conclusions?
- (4) For readers primarily interested in assessing the ON/OFF state of thick filaments, could the authors list the specific advantages of polarized second harmonic generation (pSHG)

microscopy compared to X-ray diffraction?

(5) Given that many data points were derived from the same fiber or myocyte, how did the authors address the risk of type I errors due to non-independence of measurements? Was a nested or hierarchical statistical approach used?

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Reviewer #2 (Public review):

Summary:

In striated muscle, myosin motors can dynamically switch between an energy-conserving OFF state and an activated ON state. This switching is important for meeting the body's needs under different physiological conditions, and previous studies have shown that disease-causing mutations associated with cardiomyopathies can affect the population of these states, leading to aberrant contractility. Studying these structural states in muscle has previously only been possible via X-ray diffraction, which requires access to a beam line. Here, Arecchi et al. demonstrate that polarized second-harmonic generation microscopy (pSGH), a technique that is more accessible, can be used to probe the ON/OFF states of myosin in both permeabilized and intact muscle.

Strengths:

- (1) There is an outstanding need in the field to better understand the regulation of the ON/OFF states of myosin. Currently, this is studied using X-ray diffraction, meaning that it is accessible to only a few labs. The authors demonstrate that pSGH can be used to probe the ON/OFF states of myosin both in intact and permeabilized muscle. This is a significant advance, since it makes it possible to study these states in a standard research laboratory.
- (2) The authors demonstrate that this approach can be employed in both skeletal and cardiac muscle. Importantly, it works with both porcine and mouse cardiac muscle, which are two of the most important animal models for preclinical studies.
- (3) The authors manipulate the ON/OFF equilibrium using both drugs and a genetic model of hypertrophic cardiomyopathy that has been shown to modulate the ON/OFF equilibrium. Their results generally agree with previous studies conducted using X-ray diffraction as well as biochemical measurements of myosin autoinhibition.

Weaknesses:

- (1) While the application of pSGH to the ON/OFF equilibrium is an important advance, there are limited new biological insights since the perturbations used here have been extensively characterized in previous studies.
- (2) SGH has previously been applied to study the nucleotide-dependent orientation of myosin motors in the sarcomere (PMID: 20385845). The authors have previously interpreted the value of γ as being a readout of lever arm position, but here, it is interpreted as a measure of ON/OFF equilibrium. When this technique is applied to intact muscle, it is not clear how to deconvolve the contributions of lever arm angle from the ON/OFF population (especially where there is a mix of states that give rise to the γ value). This is an important limitation that is not discussed in the manuscript.
- (3) The R403Q mutation has previously been shown to cause an increase in ATP usage. Here, the authors measure an elevated basal ATPase rate under relaxing conditions, and they interpret this as showing increased myosin ATPase activity intrinsic to the motors; however, care should be used in interpreting these results. Work from the Spudich lab has shown that

the R403Q mutation can appear as increasing motor function in some assays but depressing motor function in others (see PMID: 32284968, 26601291). Moreover, the actin-activated ATPase rate is an order of magnitude higher than the basal ATPase rate, and thus, small changes in the basal ATPase rate are unlikely to be important for physiology.

(4) The authors interpret some of their data based on the assumption that the high concentrations of drugs cause the myosin to either adopt 100% OFF or ON states. This assumption is not validated, limiting the ability to interpret the fraction of myosins in the ON/OFF states.

(5) The ATPase measurements are innovative but hard to interpret. dATP and ATP do not have identical ATPase kinetics, meaning that it is hard to deconvolve whether the elevated ATPase rate with dATP is due to changes in the ON/OFF population and/or intrinsic ATPase activity. Similarly, mavacamten reduces the rate of phosphate release from myosin, and this effect is not strictly coupled to the formation of the OFF state (e.g., see PMID: 40118457). As such, it is difficult to deconvolve drug-based changes in the inherent ATPase kinetics of the myosin from changes in the OFF-state population.

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Reviewer #3 (Public review):

Summary:

This is a very interesting paper extending the use of SHG to the study of relaxed muscle and its use to assess the order-disorder (and on /off) states of myosin heads in the thick filament. The work convincingly shows that SHG and the parameter gamma provide a reliable measure of the state of the myosin heads in a range of different relaxed muscle fibres, both intact and skinned, and in myofibrils. In mini pig cardiac fibres, the use of dATP and mavacamten increased or decreased the number of heads in the disordered state, respectively. On the assumption that these treatments push myosins fully into the disordered or ordered state, then this allows the fraction of ordered heads to be assessed under a wide variety of conditions. It is unfortunate that dATP treatment was not used (as mavacamten was) on rabbit psoas and mouse samples to further test this hypothesis.

The results with the myosin mutant R403Q support the idea that this mutation reduces the fraction of myosin heads in the ordered state and that mavacamten can recover the WT situation.

The results from SHG were compared with parallel studies using X-rays to validate the conclusions. Independent fibre ATPase data further support the conclusions.

The work is solid and provides a novel approach to assessing the activity state of muscle thick filaments. The authors point out some of the potential uses of this approach in the future, including time-resolved SHG measurements. Indeed, jumps in mavacamten or dATP concentration with time-resolved SHG could measure the rates of entry and exit from the ordered, off state of the filament. A measurement is urgently needed in the field.

Strengths:

(1) The SHG signal is convincingly shown to assess the fraction of ordered/disordered myosin heads in the thick filament of a variety of muscle fibres.

(2) The results are similar for rabbit psoas, mouse, and minipig cardiac fibres. Skinning the fibres and production of myofibrils do not change the SHG signal.

(3) Use of myosin R403Q mutant in mini pig confirms a loss of ordered myosin heads, and the ordered heads can be recovered by mavacamten.

(4) Parallel X-ray scattering and ATPase data support the conclusions.

(5) Assuming that dATP and mavacamten generate 100% disordered vs ordered myosin heads respectively, then the percentage of ordered heads can be calculated for a variety of conditions.

Weaknesses:

(1) Issues like the effect of fibre disarray and lattice spacing on the SHG signal are not well defined.

(2) The, now well-defined heterogeneity of thick filament structure is not acknowledged.

(3) dATP was only used on minipig cardiac fibres. The effect of dATP on rabbit psoas and mouse cardiac fibres would be a useful comparison and would help validate the calculation of % ordered heads.

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